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INSTITUTE OF TECHNOLOGY
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OSSP Individual Project

Installation of macOS Sonoma in Virtual Environment (Oracle VM VirtualBox)

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Introduction

Operating systems (OSs) can be referred to as software products that control computer hardware components and offer their resources to other users' applications. Operating systems vary depending on the needs and functions. For example, there is Windows used for everyday personal computing, Linux that powers many servers and developers, and macOS used in most of the computers manufactured by Apple Inc. One of the latest versions of this OS is Sonoma.

In the contemporary computing environment, virtualization can be defined as the technology that allows running several operating systems in one computer at once with the help of virtual machines. This tool makes it possible for users to test software, study various operating systems, and utilize computer resources effectively without having different hardware devices.

The purpose of the current project was to install the newest version of the macOS, namely Sonoma, in a virtual machine on a Windows laptop using Oracle VM VirtualBox software.

Objectives

The primary goal of this project is for students to familiarize themselves with the theoretical and practical aspects of installing an operating system within a virtual machine through the installation of macOS Sonoma on Oracle VM VirtualBox. During this project, students will be able to learn how to create and configure a virtual machine, allocate resources, install and configure operating system components, and complete the installation process of a modern operating system.

Another goal of this project is to examine the architecture and functionality of macOS Sonoma as a modern UNIX-based operating system. The macOS Sonoma is created by Apple and features advanced security mechanisms, improved GUI, effective memory management, enhanced multitasking abilities, and compatibility with Apple devices and services. Through this project, the student will be able to see how macOS Sonoma differs from other operating systems such as Windows and Linux in terms of usability, file system, security, and performance.

Another goal of this project is to acquire knowledge about the hardware and software needs associated with virtualization. Before setting up macOS Sonoma on VirtualBox, it is important to comprehend the function of RAM allocation, number of processors, disk space, EFI boot mode, availability of virtualization in BIOS/UEFI, and ISO installation files. This is crucial since having not enough hardware may cause the installation process to fail or make your system perform poorly. Thus, the

project will allow students to acquire knowledge about managing hardware resources in guest and host environments.

Moreover, another task of this project is to detect the issues which may occur when working on virtual machines and learning how to overcome them. While installing macOS Sonoma, one may face issues including but not limited to “No bootable medium found,” black screen, startup problems, CPU settings not supported, lack of ISO files, or storage issues. Working with problems in the mentioned order will help you enhance your analytic thinking and practical skills.

Finally, the task of introducing modern filesystems employed in operating systems will be completed as well. In particular, macOS Sonoma employs Apple File System (APFS), which is designed for SSD media, snapshot support, encryption, and high-performance features. By studying other filesystems, including NTFS, FAT32, exFAT, ext4, Btrfs, and ZFS, students will be able to grasp the reasons why different OSes employ different filesystems due to design objectives, security issues, etc.

Moreover, the project will address the subject of virtualization, namely its essence, purposes, and significance in today's computing environment. This concept is broadly used in teaching, software engineering, cloud computing, cybersecurity lab testing, and enterprise-level virtual data centers. Using VirtualBox, learners will have an opportunity to see how virtualization works by learning what role a hypervisor plays in abstracting hardware resources and running guest OSes on top of the host machine.

Finally, the project will provide information about standardization and implementation of UNIX. Since macOS is an OS built according to the UNIX model, it makes sense to study ISO C, IEEE POSIX, FIPS, BSD systems, Solaris, and Mac OS X.

Additionally, this assignment will develop the skills of documenting and reporting information. Documenting the steps of installation, screenshotting, listing of problems and solutions and presentation of results in the form of a structured report is a skillful task that is valuable for software engineers and system administrators. Well-done documentation enables other people to repeat your actions and resolve their issues.

Lastly, the general goal of this assignment is combining theory with practice. Instead of theoretical studying of operating systems, the student acquires experience in working with virtualization programs, installing an operating system, working within its environment and resolving actual technical problems.

Requirements

Installing macOS Sonoma on a virtual machine requires correct setup of hardware and software resources. Because virtualization enables running one OS within another OS, the computer hardware will have to meet certain requirements to enable both operating systems to operate simultaneously. Otherwise, the installation might result in failure, sluggishness, or instability. For that reason, requirement verification prior to the start of work is vital.

For this particular project, the main requirements include hardware requirements, software requirements, firmware setup requirements, and user requirements.

1. Hardware Requirements

Processor (CPU)

A modern CPU supporting virtualization is needed. The processor needs to support:

Intel VT-x** for Intel CPUs

AMD-V** for AMD CPUs

These features enhance the performance of virtualized systems and provide compatibility with guest OSes like macOS Sonoma.

Recommended CPUs:

* Intel Core i5 / i7 / i9

* AMD Ryzen 5 / Ryzen 7 / Ryzen 9

It is recommended to choose a CPU featuring multiple cores, as several cores will be dedicated to the guest operating system, while the remaining cores will continue to run the host OS.

Random Access Memory (RAM)

RAM is required by both the host system and the virtual machine.

Total System RAM Requirement:

Minimum - 8 GB

Recommendation - 16 GB or higher

Recommendation of the amount of allocated RAM for macOS Sonoma VM:

8 GB (8192 MB)

Lack of RAM can result in a slowdown or freeze of the host and guest systems.

Storage Capacity Requirements

Storage requirements include:

Installation of VirtualBox
macOS Sonoma ISO image
Virtual Hard Drive
Data of the user and updates

Free space requirement:

Minimum - 80 GB

Recommendation - 120 GB or higher

Highly recommended to use SSD as it has significantly faster startup speed compared to HDD.

Graphics Processing Unit

It can be helpful when it comes to fast rendering of the display.

Some of them:

Intel UHD / Iris Graphics
NVIDIA Graphics
AMD Radeon Graphics

Amount of VirtualBox video memory:

128 MB

Internet Connection

Internet connection is required to:

Install VirtualBox

Install macOS Sonoma

Read the documentation
Make updates and use online services

2. Software Requirements

Host Operating System

The computer on which the virtual machine will be hosted should have the following installed:

Windows 10

Windows 11

This project utilizes Windows as the host operating system.

Virtualization Software

The software that will be used for the creation and operation of the virtual machine will be:

Oracle VM VirtualBox

VirtualBox will enable the user to perform the following:

Create virtual machines

Assign RAM and CPU

Connect ISO images

Mimic hardware components

Install guest operating systems

macOS Sonoma Installer Image File

An installation file is necessary to install macOS Sonoma, such as:

ISO file

Recovery image

Bootable image

Without an installation file, the virtual machine cannot operate.

Utility Software (Optional)

Some useful applications include:

WinRAR / 7-Zip (uncompressed files)

Screenshot utility for documentation purposes

Text editor for note-taking

3. BIOS/UEFI Requirements

The firmware settings of the host computer must enable virtualization.

Required settings:

Intel VT-x = Enabled

AMD-V = Enabled

Optional settings if required:

Secure Boot = Disabled (only if conflicts arise)

If virtualization is not enabled, VirtualBox may encounter issues when launching 64-bit operating systems.

4. Recommended Virtual Machine Settings

For proper installation of macOS Sonoma, the recommended virtual machine settings are:

Virtual Machine Name: macOS Sonoma

Memory: 8192 MB

Number of Processor Cores: 4

Video Memory: 128 MB

Storage Space: 120 GB

EFI Option Enabled

SATA Controller Option Enabled

This setting is a perfect combination of performance and stability.

5. User Requirements

The user will be expected to have:

- Knowledge of computers
 - Experience installing software
 - Patience when dealing with issues that may arise
 - Ability to follow instructions
 - Ability to take screen captures
6. Reasons Why the Requirements are Necessary

Following the requirements leads to:

- Fast installation process
- Good performance
- Minimal crashes
- Great experience using virtual machines
- Project success

Failure to meet the requirements will lead to:

- Low performance
- Boot failures
- Installation problems
- Freezing of VMs

Conclusion

User requirements play an essential role in the success of any virtualization and operating system installation. For one to install macOS Sonoma on a virtual machine, there must be sufficient computing power, enough memory, adequate space, and virtualization capabilities on the computer.

Hardware

This term describes physical computer components for running software and the operating system. The hardware plays an essential role in this project because it will be used for running the host operating system (Windows) and the guest operating system (macOS Sonoma) using VirtualBox simultaneously.

- Primary Hardware Specifications
- Processor (CPU)

It is necessary to have a new processor with a virtualization feature.

Examples:

Intel Core i5/i7

AMD Ryzen 5/7

The processor must have the following capabilities:

Intel VT-x

AMD-V

These characteristics facilitate correct work with virtual machines within VirtualBox.

RAM (Memory)

It is crucial to have enough memory for Windows and macOS Sonoma.

Minimum: 8 GB

Recommended: 16 GB

For the virtual machine:

8 GB RAM (recommended)

Storage

Space on a hard drive is needed for VirtualBox installation, files of the macOS operating system, and virtual disk.

Minimum: 80 GB

Recommended: 120 GB

SSD storage is preferred to HDD because it is faster.

Graphics

An average graphics card is sufficient for the display.

Recommended VirtualBox video memory:

128 MB

Internet Access

An internet connection is necessary for downloading software, installer files, and updates.

Hardware Used in This Project

Multicore processor

16 GB RAM

120 GB virtual disk

Software

The term software describes the application needed for the installation and execution of macOS Sonoma in the virtual machine.

Software Needed

Windows 10/11 – Operating system

Oracle VM VirtualBox – Virtualization software

macOS Sonoma ISO – Installation program

7-Zip / WinRAR – Decompression application (optional)

Purpose

This collection of applications serves to establish the virtual machine and operate the installation program of macOS Sonoma.

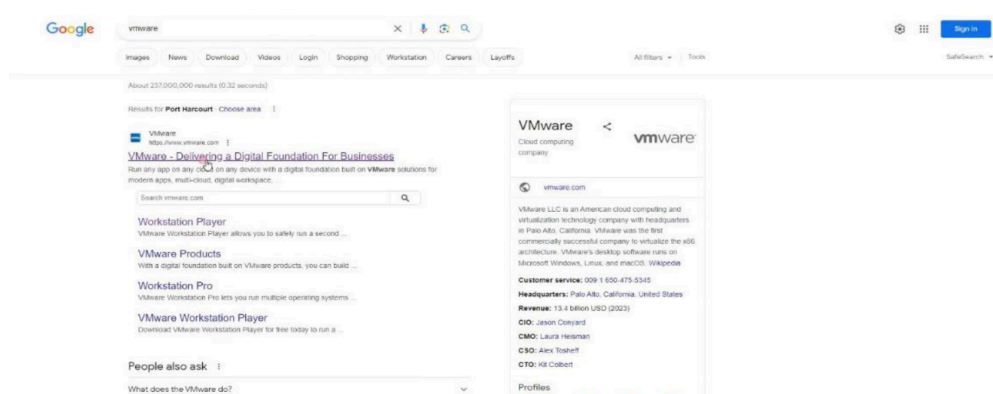
Conclusion

Appropriate software is needed for the installation and operation of the virtual machine.

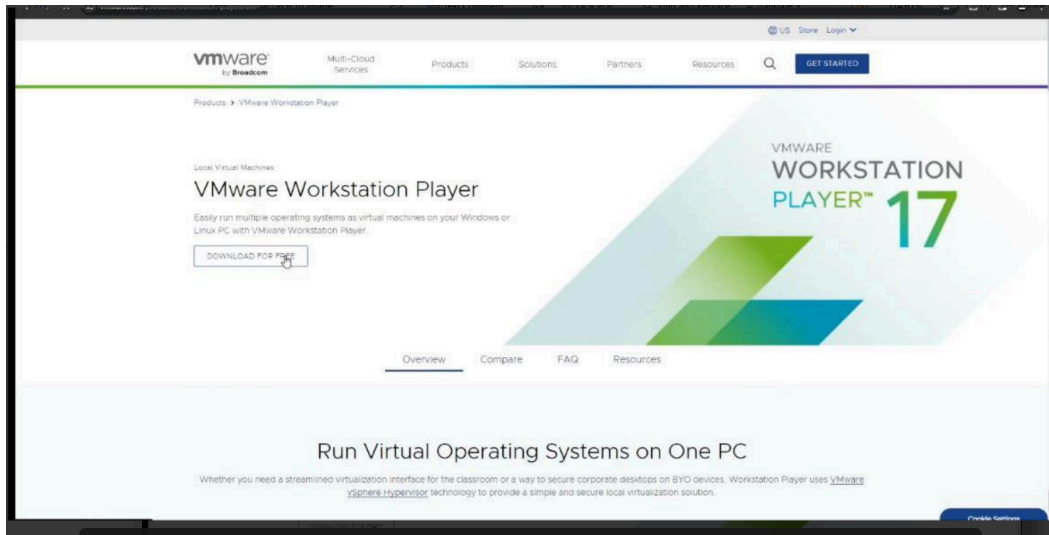
Installation steps

To install the latest version of Mac OS Sonoma on VMware :

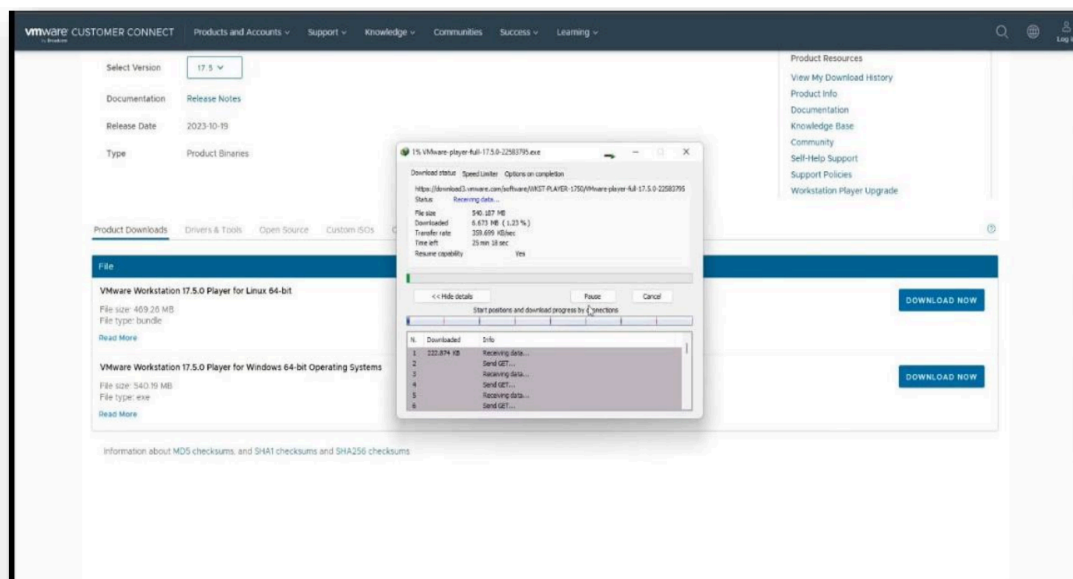
❖ The first thing that we have to do is downloading the required files so head to our google



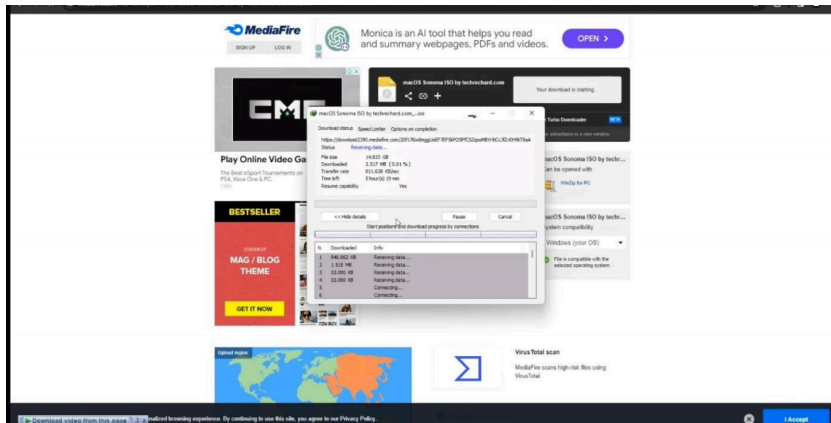
- ❖ First we have to download VMware ,and search for VMware Click on the link to VMware.com, then click on products, and then select VMware workstation player under desktop hypervisor.
- ❖ Click on download for free, and over here we can see the version of VMware available, now click on go to downloads at the bottom.



- ❖ Here you can see VMware for Linux and for windows so we're going to download for windows since we're using a windows computer, so click on download now under windows and it should start downloading.



❖ Next after downloading VMware we also have to download Mac OS ISOfile which we going to use for the installation. clicking on the downloadbutton and it should start downloading.



The file size is about 14.8 gig, it would recommend we use n internet download manager if our internet is not stable.

❖ Lastly we have to download Auto un locker for VMware so let's open a new

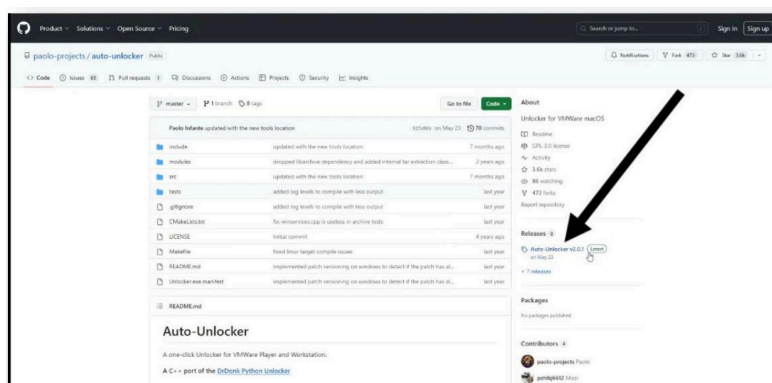
tab and then search for auto un locker for VMware ,

❖ Now we can see polo projects Auto un locker, and let's click on it.

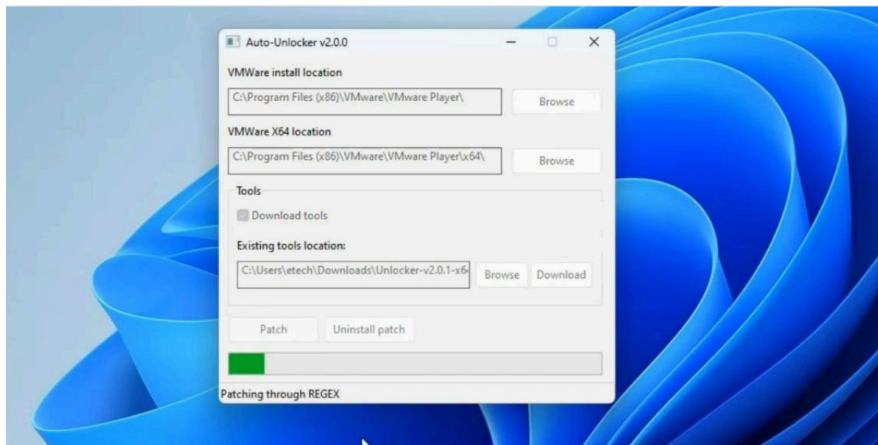
❖ Now this two is going to allow us to run Mac OS on our VMware smoothly.

❖ Look by the right and we would see the latest release, which we're going to

click on and then scroll down and then and we should see un locker and download the zip



- ❖ So after downloading VMware mac OS Sonoma ISO file and unlocker we can close our browser and head to download.
- ❖ Then in downloads ,we found VMware ,unlocker ,and macOS ISO file .
- ❖ First we have to install VMware by clicking it.



- After installing VMware now we have to run unlocker so let's head back to our download folder and then extract unlocker
- Now let's open the extracted folder, right click and click on run as administrator.
- Now let's click on patch ,it's going to apply patch to VMware so we can run mac OS Sonoma, so let's click on patch this should take a while to download the required files and run the patch.
- After we are done with the patch ,now click on ok and then we can close this.
- now let's open VMware from our desktop, and then create a new virtual Machine.



❖ We're going to select installer disk image file and then click on browse now

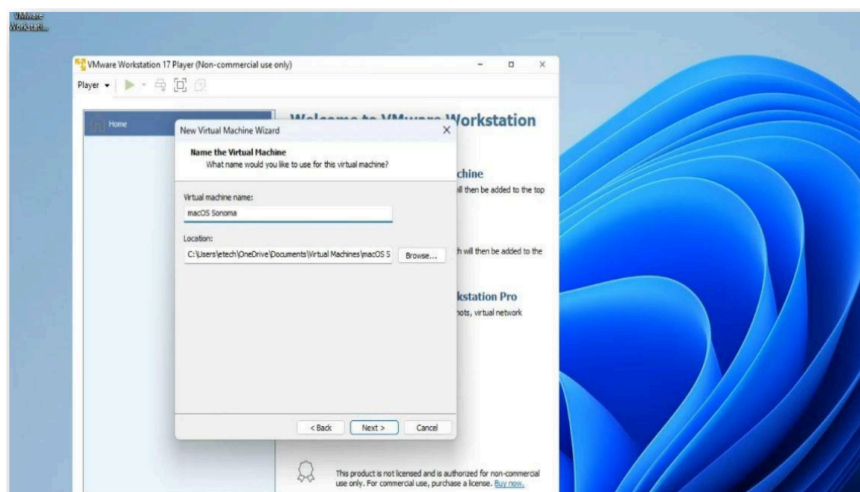
we select our Mac OS Sonoma ISO file from our download folder ,open, and then next...,now this was the reason we used auto unlocker patch if we don't run that we won't get an option for Mac OS X.

❖ Now we set the version to Mac OS 14 and then enter our machine name,

then I'm going to call this Mac OS Sonoma, and we're going to leave the location where to install as it is.

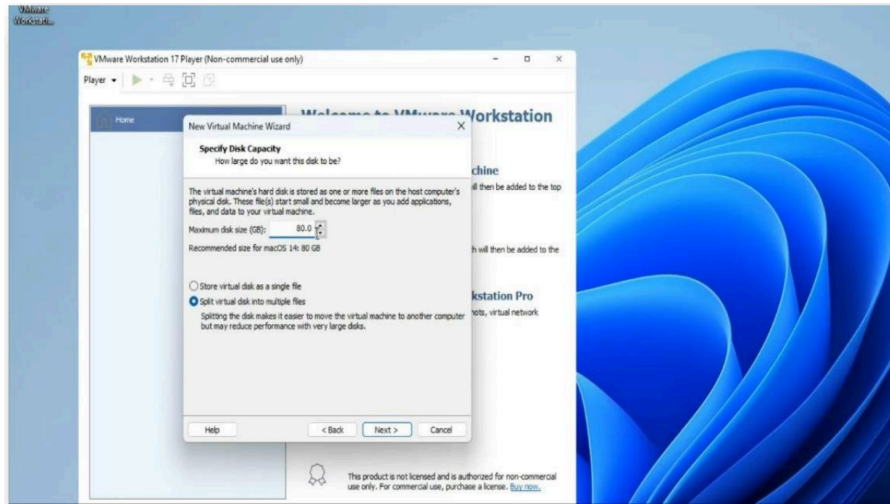
❖ Now we set the version to Mac OS 14 and then enter our machine name

,then I'm going to say it Mac OS Sonoma and we are going to set the location to install as it is.

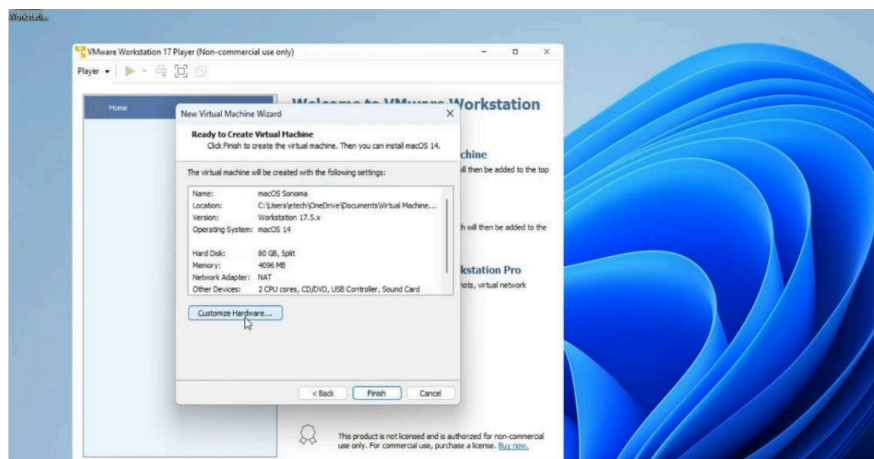


❖ Then set a size for our installation,it is recommended any thing above 80GB,so I just leave it on 80GB.

We leave it at split virtual disk in to multiple file so if we want to transfer it into another system it will be easy.

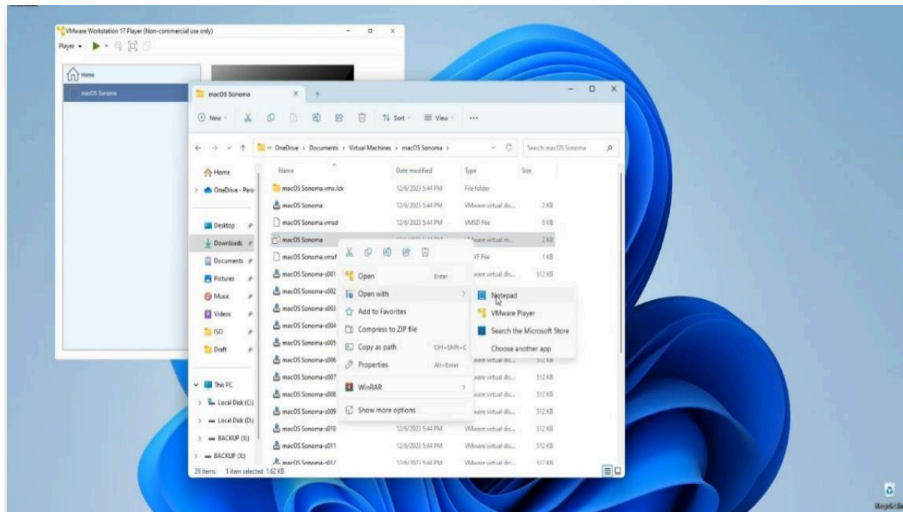


❖ Click on next and click on customize Hardware ,so we can make few changes .

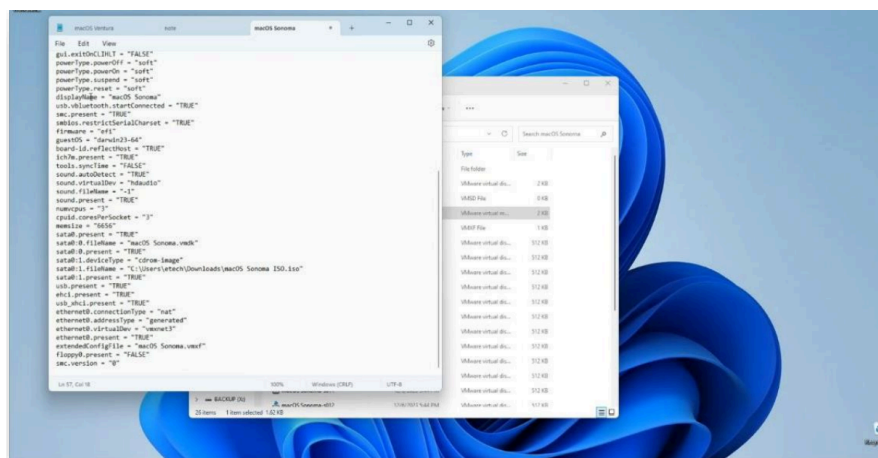


- We can increase the RAM to whatever we want, now in my case I'm going to give it about 6.5GB, we shouldn't allocate more than 50% of our RAM to our virtual machine else it will slow down our computer.
- Same for processor as well, I'm going to give it 3 CES and then everything else we can leave as it is, and then we click on close ,and click on finish.

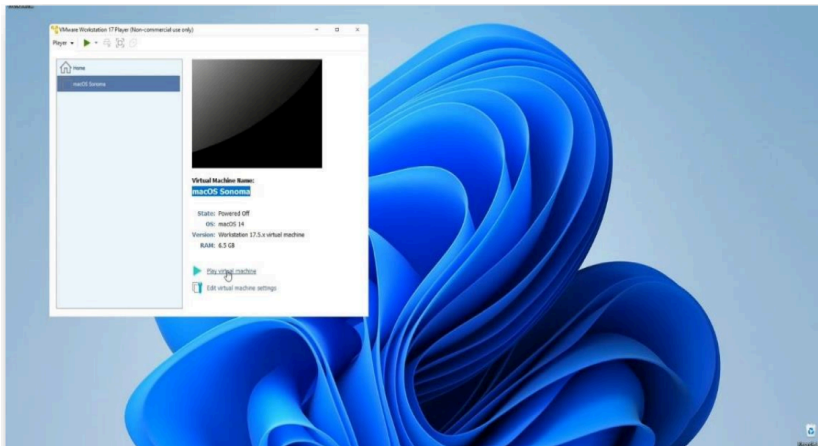
- And we are almost to start the installation, next click on edit virtual machine settings, so we can get the directory where this machine is going to be running from.
- Go to the hard disk and copy the disk file, then we need to copy it to place where it says the name of our virtual machine which is Mac OS Sonoma in my case.



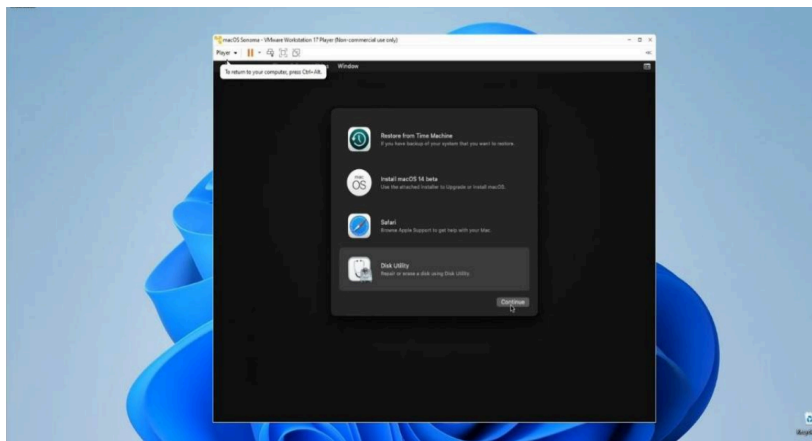
- Then we have to edit the configuration file and if we look closely we would see a file with a type called VMware virtual machine configuration. right click on that file and then open with notepad. We have to make changes here, and scroll down and write "smc.version="0"", and go to file and save it. then close notepad.



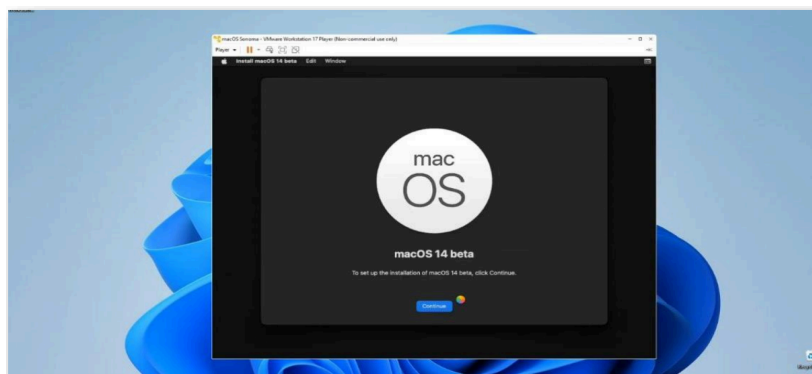
- Now we can start our virtual machine and begin the installation .click on play virtual machine.



- Now set language for installation and then ,first we have erased the disk we assigned and then install Mac OS.

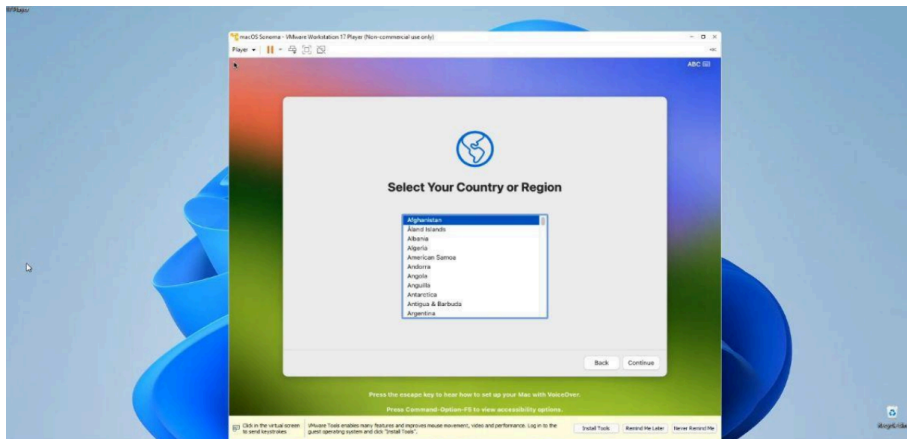


- Go to disk utility ,then we can see under internal drives VMware virtual SATA hard drive ,and I'm going to call it Mac OS 14.

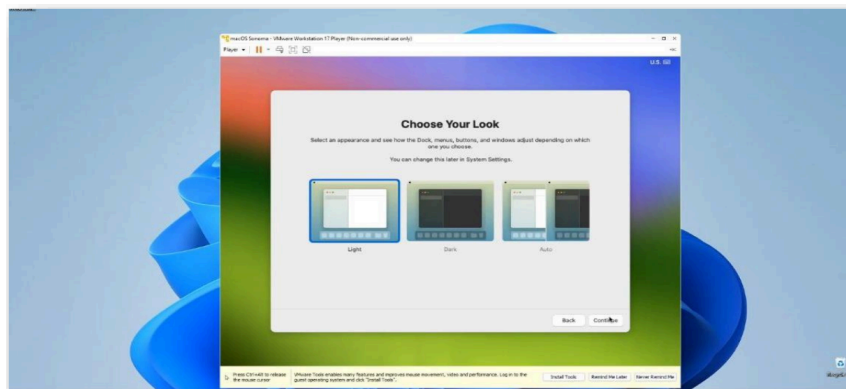


- click on(continue...continue...and agree to the terms)and then select our device and click continue .And it should begin installin ,it

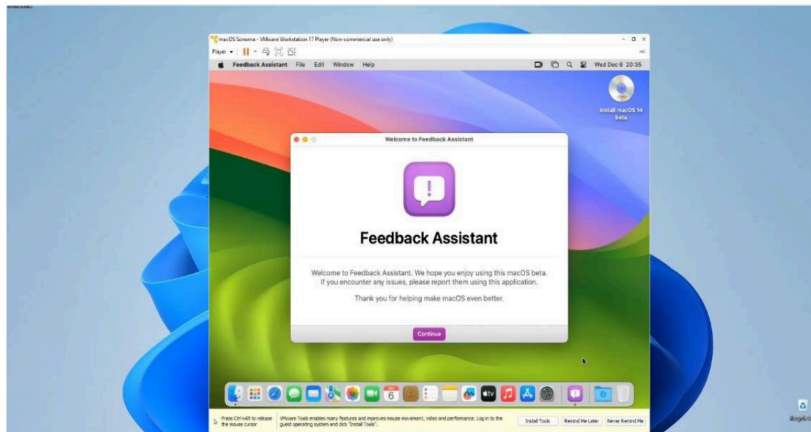
will take about an hour to install or it may restart severally while installing.



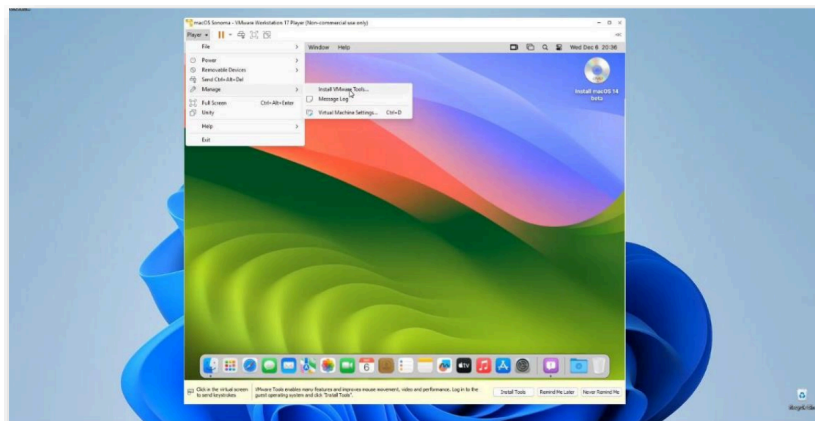
- Installation is completed .now we can set up based on our personal preference, enter username and password which we going to access our machine .

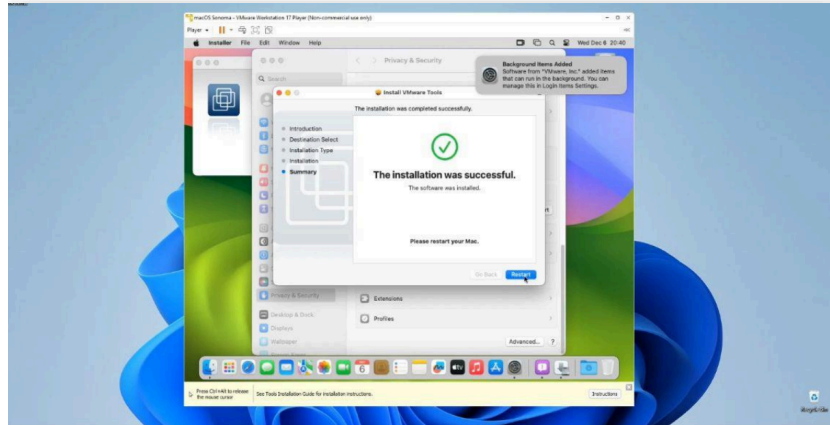


- Here we can choose between light mood or dark mood ,or we can also change this later in settings and we are done with the installation.



- There are few issues we need to take in to consideration :
 - Our machine is running quite slow ,that is because of graphics so we click on about this Mac, here we can increase the graphics in which is given in megabyte.
 - The other issue is ,if we want to resize the machine's doesn't work, and to view it in full screen mode still doesn't work.
 - To fix that in graphics issue we need to install VMware tools and to install that we just have to go to player and then click on manage and click install VMware tools .
 - Restart our PC, then use our password to sign in again. No the graphics has increased to 128mega byte and we could also drag the screen and make it full screen. now everything is working without any issue.





- All the above are the installation processes that we need to follow in order to install Mac OS Sonoma on our PC

Issues (Problems Faced) and Their Solutions

Various problems may emerge while installing or working with macOS Sonoma in the virtualized environment. The knowledge of the issues encountered and the means for overcoming such difficulties will help users to install the OS correctly and become better at dealing with potential complications.

1. Installation Is Failed or Freezing

Problem description:

The installation process is interrupted, hangs, or freezes for an extended period.

Possible solution:

Try starting the virtual machine once more and repeating the installation process.

- * Make sure there is sufficient free disk space (it should be 20 GB or higher).
- * Allocate more RAM and CPU power on VirtualBox.
- * Inspect the ISO file to verify its integrity.

2. Incompatibility with Some Programs

Problem description:

The applications used before the installation do not work anymore and keep crashing upon opening.

Possible solution:

- * Update all software to the latest versions.
- * Install only those software programs that support macOS Sonoma.
- * Uninstall any outdated software.

3. Boot Loop

Problem description:

The operating system cannot start successfully and is constantly restarting.

Possible solution:

- * Restart the virtual machine and launch the system again.
- * Inspect the EFI settings in VirtualBox.
- * Consider reinstalling macOS if boot files were corrupted.
- * Repair the virtual hard drive using Disk Utility.

4. Problems with Wi-Fi or Internet

Symptoms:

There is no internet connection available inside the virtual machine.

Solutions:

- * Check virtual network configuration.
- * Use NAT or bridged adapter mode.
- * Restart both host machine and virtual machine.
- * Reconnect to the network.

5. Performance Problem

Symptoms:

Computer runs slow or laggy.

Solutions:

- * Allocate more memory to the VM.
- * Give more processing power to the virtual machine.
- * Close other running programs on the host system.
- * Use an SSD drive for better performance.

6. File Backup or Migration Problem

Symptoms:

Cannot migrate or transfer files or backup data.

Solutions:

- * Ensure that the backup files can be accessed.
- * Utilize shared folder in VirtualBox.
- * Try migrating after completing the installation process.

7. Specific Error Message During Installation

Symptoms:

An installation error message appears.

Solutions:

- * Pay close attention to the error message.
- * Verify system requirements.
- * Have an authentic installer file.
- * Look up the error code online in the official support.

8. External Devices Do Not Work

Symptoms:

USB devices, scanner, or printer does not connect.

Solutions:

- * Make sure USB support is enabled.
- * Download the latest driver updates, if any.
- * Unplug and plug again the external device.

9. User Account Problems

Indicators:

Cannot access your user account or the user files cannot be found.

Possible Solutions:

- * Restart the system and attempt again.
- * Boot the computer in Safe Mode.
- * Determine if files are saved under other user accounts.

10. System Preferences Crashes

Indicators:

System Preferences application does not work.

Possible Solutions:

- * Restart the computer.
- * Update to macOS Sonoma version.
- * Delete damaged preference files.

General Guidelines

- * Ensure that your hardware is compatible with the operating system.
- * Create backups for essential files.
- * Install using legitimate software resources.
- * Provide sufficient memory, processor, and hard drive space.
- * Expect the first-time setup process to be slow due to initial configuration.

Conclusion

Though many problems could arise while installing macOS Sonoma, most of them could be overcome through proper setup processes and resource allocation.

File System Supports

A file system is the method used by an operating system to store, organize, manage, and retrieve data on storage devices such as hard disks, SSDs, and USB drives. macOS Sonoma primarily uses **APFS (Apple File System)**, which is the default file system for modern Apple devices. APFS replaced the older **HFS+** file system because it offers better speed, security, and reliability.

APFS is specially designed for modern storage devices such as SSDs and flash drives, making it the best choice for macOS Sonoma.

Main File Systems Supported by macOS Sonoma

- **APFS** – Default file system for macOS Sonoma
- **HFS+** – Older Mac file system
- **FAT32** – Used for USB drives and compatibility with Windows
- **exFAT** – Best for sharing files between Windows and macOS
- **NTFS** – Read-only support by default in macOS

Why APFS is Preferred in macOS Sonoma

1. High Performance

APFS is optimized for speed and efficiency, especially on SSD storage devices. It allows faster booting, quick file access, and smoother system performance compared to HFS+.

2. Space Efficiency

APFS uses **copy-on-write** technology. When a file is changed, the system creates new metadata instead of rewriting the entire file. This saves space and reduces fragmentation.

3. Snapshots

APFS supports snapshots, which save the state of the system at a certain time. This is useful for:

- Backup recovery
- Restoring deleted files
- Returning to previous system states

4. Strong Security and Encryption

APFS provides built-in encryption features that help protect personal and sensitive data. Users can encrypt entire drives or selected volumes.

5. Reliability

APFS improves data integrity using metadata protection and crash-safe design. This reduces the chance of file corruption.

6. File and Folder Cloning

APFS can clone files instantly without copying all data physically. This saves both storage space and time.

7. Multi-Volume Support

Multiple volumes can exist inside one APFS container and share free space dynamically. This allows better storage management.

Comparison with Other File Systems

File System	Used In	Features
-------------	---------	----------

APFS	macOS Sonoma	Fast, secure, snapshots
HFS+	Older macOS	Legacy support
FAT32	USB drives	High compatibility
exFAT	Windows/macOS sharing	Supports large files
NTFS	Windows	Read-only in macOS

Conclusion

macOS Sonoma mainly supports APFS because it is faster, more secure, and more reliable than older file systems. It is designed for modern SSD storage and provides advanced features such as encryption, snapshots, cloning, and efficient space management. Therefore, APFS is the most suitable file system for macOS Sonoma.

Advantages and Disadvantages of macOS Sonoma

macOS Sonoma is a modern operating system developed by Apple. It provides improved performance, security, and user experience. However, like any operating system, it has both advantages and disadvantages.

Advantages of macOS Sonoma

1. Better Performance

macOS Sonoma is optimized for newer hardware, giving faster speed, smoother multitasking, and better overall system performance.

2. Improved User Interface

It provides a modern and attractive interface with better design, making the system easier and more enjoyable to use.

3. Desktop Widgets

Users can place widgets directly on the desktop for quick access to weather, calendar, reminders, and other information.

4. Strong Privacy and Security

macOS Sonoma includes advanced privacy controls, secure permissions, and data protection features.

5. Safari Improvements

The Safari browser has better speed, tab management, and stronger privacy protection for safer browsing.

6. Better Gaming Experience

Features such as **Game Mode** improve gaming performance and reduce lag.

7. Apple Device Integration

macOS works smoothly with iPhone, iPad, and other Apple devices through AirDrop, iCloud, Handoff, and Continuity features.

8. Improved Video Conferencing

Better screen sharing, Presenter Overlay, and camera tools help with online meetings and remote work.

Disadvantages of macOS Sonoma

1. Compatibility Issues

Some older applications, drivers, or devices may not work properly after upgrading.

2. High Hardware Requirements

Sonoma performs best on newer Mac devices, so older systems may run slowly.

3. Learning Curve

Users coming from Windows or older macOS versions may need time to learn new features.

4. Bugs in Early Versions

New operating systems sometimes release with bugs or stability issues.

5. High Resource Usage

Some modern features use more RAM, CPU, and storage resources.

6. Limited Customization

Compared with Windows or Linux, macOS offers fewer customization options.

7. Expensive Ecosystem

macOS officially runs only on Apple hardware, which is often more expensive.

macOS Sonoma offers excellent performance, security, and integration with Apple devices. It is ideal for productivity and modern computing. However, it may have compatibility issues, higher hardware requirements, and limited customization for some users.

Conclusion

MacOS Sonoma is an up-to-date operating system that enhances the Mac performance in terms of security, efficiency, usability, and beauty. This operating system incorporates a variety of functions that provide enhanced performance, greater ease of use, and versatility for professional purposes.

One of the most important advantages of macOS Sonoma is related to personalization. This system offers beautiful new wallpapers and

personalized widgets that can be placed directly on the desktop providing users with quick access to weather, calendar notifications, reminders, and other helpful details.

Another aspect where macOS Sonoma stands out is improved communication and remote work opportunities provided by video conferencing tools. The new features include Presenter Overlay, screen sharing enhancement, and visual reactions that ensure smooth interaction during video conferences.

Moreover, macOS Sonoma enhances users' productivity with advanced Focus Mode features designed to limit distractions during studies or work.

What is more, the newest version of Safari offers improved security, increased speed, and tabs organization function.

Finally, creative apps like Photos, Music have been greatly updated as well.

Thus, macOS Sonoma is a perfect combination of aesthetics and functionality. This operating system creates a fresh, safe, and productive environment.

Future Outlook / Recommendation

Future outlooks for macOS Sonoma appear promising as Apple continuously works on improving its operating systems by means of introducing new updates and additional features to enhance performance, security, and stability. As the adoption rate increases, software developers will make sure their products are optimized to be compatible and deliver great performance.

Moreover, macOS Sonoma is likely to integrate with upcoming devices from Apple better to ensure fast processing speed, increased battery performance, and superior graphics capabilities. The high value the

company puts on security and privacy makes macOS Sonoma an appealing solution for customers who seek protection of their personal information.

More and more users will share their experiences, including writing tutorials and guides to solve common issues and assist those who are at the beginner level.

It is advised to upgrade to macOS Sonoma when possible as long as your device supports it. Prior to upgrading, users should make sure that they have a backup of all important data and that their software is compatible with it.

What is Virtualization?

Virtualization is an approach where the user can make virtual versions of different computer resources including hardware, storage, network, and operating systems. Multiple operating systems and applications are executed on a single physical computer simultaneously. This increases the efficiency of hardware usage, reduces costs, and facilitates the management process.

Modern operating systems employ virtualization extensively for cloud computing, software testing, cybersecurity, and educational purposes. Thus, the user can create several virtual computers.

Types of Virtualization

1. Hardware Virtualization

Virtualization of hardware makes it possible to create virtual machines which run their own operating systems independently. Examples of technologies which implement hardware virtualization include VirtualBox, VMware, and Hyper-V.

2. Operating System-Level Virtualization

It is also known as containerization when the user can run different isolated environments on a single operating system kernel. The main examples include Docker and LXC.

3. Storage Virtualization

Different storage drives become parts of a single logical environment in order to simplify administration.

4. Network Virtualization

Creation of independent virtual networks without using physical hardware.

Conclusion

Virtualization is one of the essential technologies in contemporary computing.

Why do we use virtualization?

Virtualization in modern operating systems is important for several reasons:

1. Resource Utilization: Virtualization allows multiple virtual machines (VMs) to

run on a single physical server, maximizing hardware resource utilization. This

leads to better performance and efficiency, as resources can be allocated

dynamically based on demand.

2. Isolation: Each VM operates in its own environment, providing strong isolation

between different workloads. This means that issues in one VM (such as crashes

or security breaches) do not affect others, enhancing security and stability.

3. Scalability: Virtualization makes it easier to scale applications and services.

New VMs can be quickly provisioned to handle increased loads, allowing organizations to respond rapidly to changing demands.

4. Simplified Management: Virtualization simplifies the management of IT

resources. Administrators can manage multiple VMs from a single interface,

automate deployment, and streamline backup and recovery processes.

5. Cost Efficiency: By consolidating workloads onto fewer physical servers,

organizations can reduce hardware costs, energy consumption, and data center

space. This leads to lower operational costs over time.

6. Testing and Development: Virtualization allows developers to create isolated

environments for testing applications without the risk of affecting production

systems. This facilitates rapid development cycles and enhances software quality.

7. Disaster Recovery: Virtual machines can be easily backed up and replicated,

making disaster recovery processes more efficient. In case of hardware failure,

VMs can be quickly restored on different physical machines.

8. Legacy Support: Virtualization enables organizations to run legacy applications that may not be compatible with modern hardware or operating systems by encapsulating them in a VM.

How Do We Use Virtualization?

Virtualization is used by creating virtual versions of computer resources such as servers, storage, networks, and operating systems. It allows multiple operating systems and applications to run on one physical computer. This helps save cost, improve efficiency, and make system management easier.

1. Hypervisors

A hypervisor is software that creates and manages virtual machines.

- **Type 1 Hypervisor:** Runs directly on hardware. Examples: VMware ESXi, Hyper-V.
- **Type 2 Hypervisor:** Runs on an existing operating system. Examples: Oracle VirtualBox, VMware Workstation.

2. Virtual Machines (VMs)

Each virtual machine acts like a separate computer with its own operating system, memory, storage, and applications.

3. Resource Management

The hypervisor shares CPU, RAM, disk, and network resources among virtual machines efficiently.

4. Isolation and Security

Each VM works independently, so problems in one VM usually do not affect others.

5. Snapshots and Cloning

Virtualization allows users to save system states (snapshots) and copy VMs quickly (cloning).

6. Common Uses

- Running multiple operating systems
- Software testing
- Server consolidation
- Cloud computing
- Backup and recovery
- Running old applications

Conclusion

Virtualization is widely used to improve hardware utilization, flexibility, and system management in modern computing.

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